INFO90007 – Social Computing

Tutorial 10 - *Evaluating Social Systems*

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Reminder

A3 Demo

18th October, 11:59 pm

Example progression of prototypes for A3:

Mid (A2) -> High1 -> Eval -> High 2

Wireframes -> Mockup1 -> *Eval* -> Mockup 2

Outline

- Usability Heuristics
- Observational methods
- Query Techniques

What is Usability?

Usability is a measure of how well

a specific user

in a specific context

can use a product/design

to achieve a defined goal

effectively, efficiently, and satisfactorily.

Dictionary

Search for a word

usability

/ˌjuːzəˈbɪləti/

noun

the degree to which something is able or fit to be used.
"it was important to measure the usability of each product"

Definitions from Oxford Languages

Feedback

Translations and more definitions

Usability Heuristics

1. Visibility of system status
2. Match between system and the real world
3. User control and freedom
4. Consistency and standards
5. Error prevention
6. Recognition rather than recall
7. Flexibility and efficiency of use
8. Aesthetic and minimalist design
9. Help users recognize, diagnose, and recover from errors
10. Help and documentation

Nielsen Norman Group

Jakob's Ten Usability Heuristics

1 Visibility *of* System Status

Designs should **keep users informed** about what is going on, through appropriate, timely feedback.

2 Match between System and the Real World

The design should speak the users' language. Use words, phrases, and concepts **familiar to the user,** rather than internal jargon.

3 User Control *and* Freedom

Users often perform actions by mistake. They **need a clearly marked "emergency exit"** to leave the unwanted state.

4 Consistency *and* Standards

Users should not have to wonder whether different words, situations, or actions mean the same thing. **Follow platform conventions.**

5 Error Prevention

Good error messages are important, but the best designs **prevent problems** from occurring in the first place.

6 Recognition Rather Than Recall

Minimize the user's memory load by making elements, actions, and options visible. Avoid making users remember information.

7 Flexibility *and* Efficiency of Use

Shortcuts — hidden from novice users — may **speed up the interaction** for the expert user.

8 Aesthetic *and* Minimalist Design

Interfaces should not contain information which is irrelevant. Every extra unit of information in an interface **competes** with the relevant units of information.

9 Recognize, Diagnose, *and* Recover from Errors

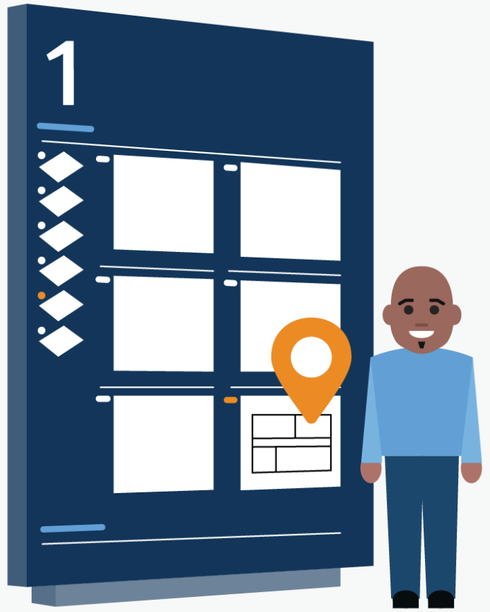
Error messages should be expressed in **plain language** (no error codes), precisely indicate the problem, and constructively suggest a solution.

10 Help *and* Documentation

It's best if the design **doesn't need** any additional explanation. However, it may be necessary to provide documentation to help users understand how to complete their tasks.

1. Visibility of system status

The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.



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1. Visibility of system status





2. Match between system & the real world

The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system-oriented terms. Follow *real-world* conventions, making information appear in a natural and logical order.



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2. Match between system & the real world



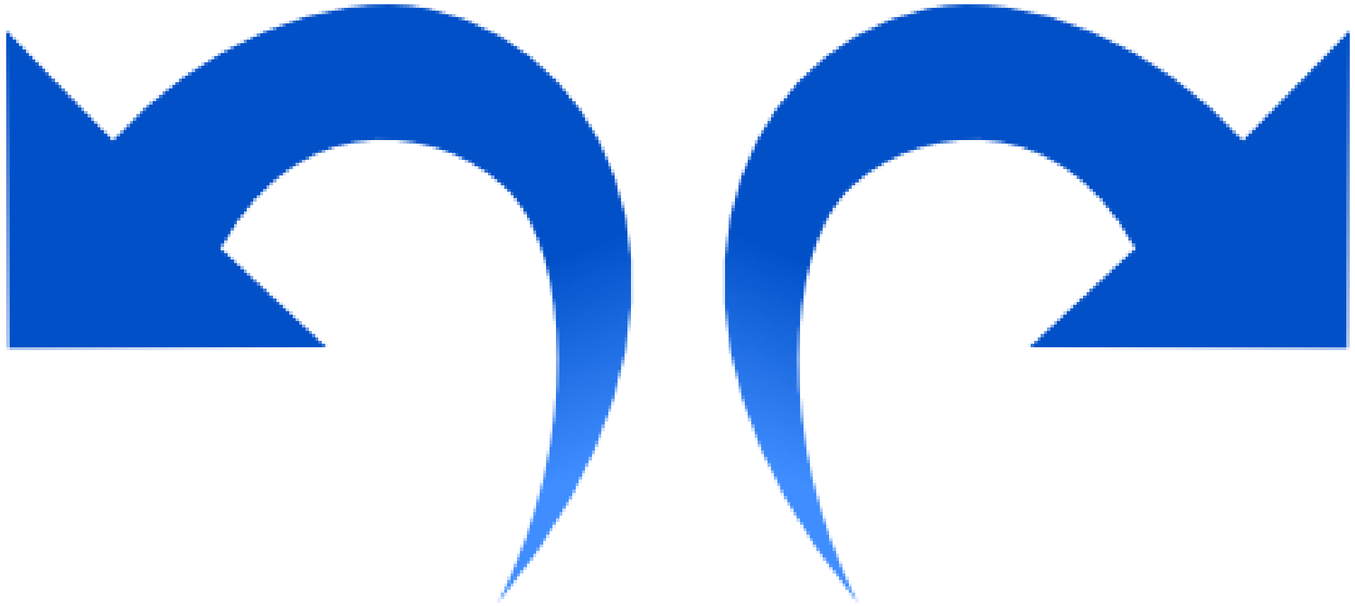
Add to Cart

3. User control and freedom

Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue.
Support undo and redo.

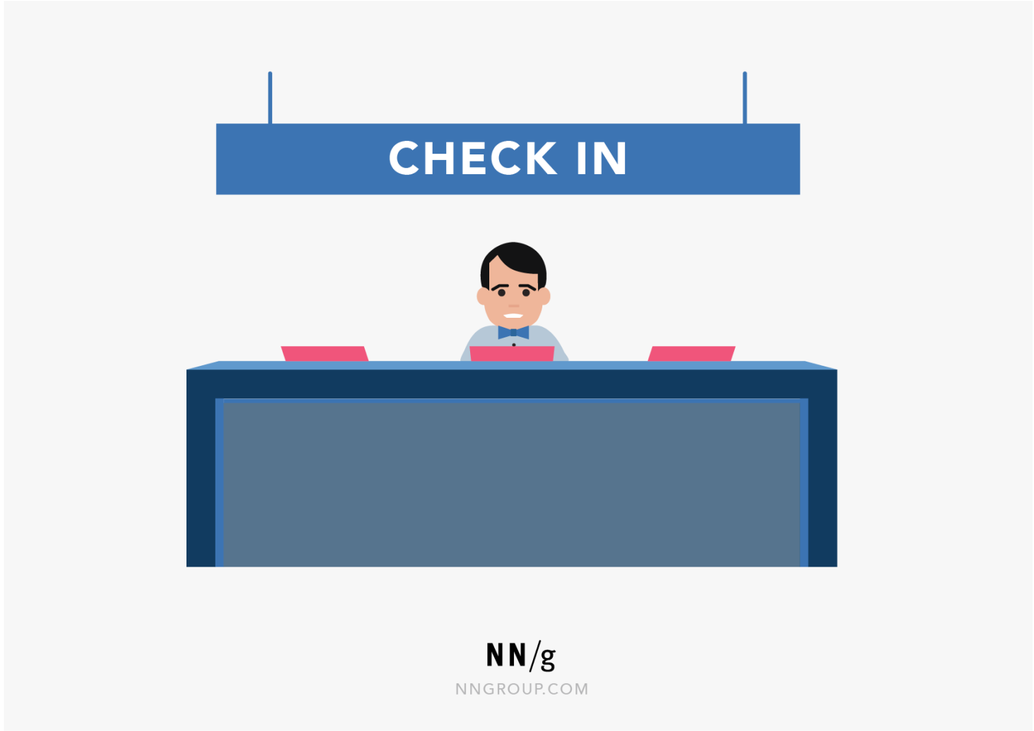


3. User control and freedom

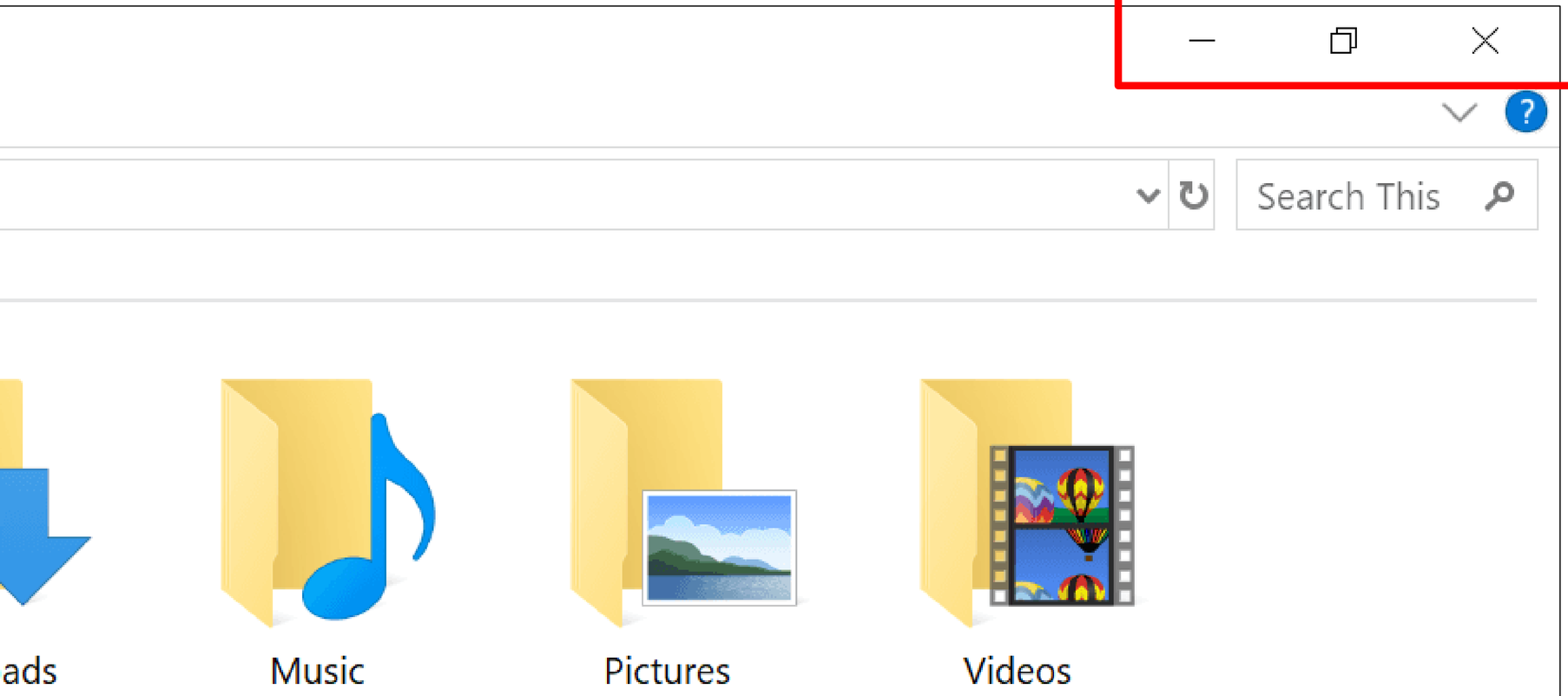


4. Consistency and standards

Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow *platform* conventions.



4. Consistency and standards

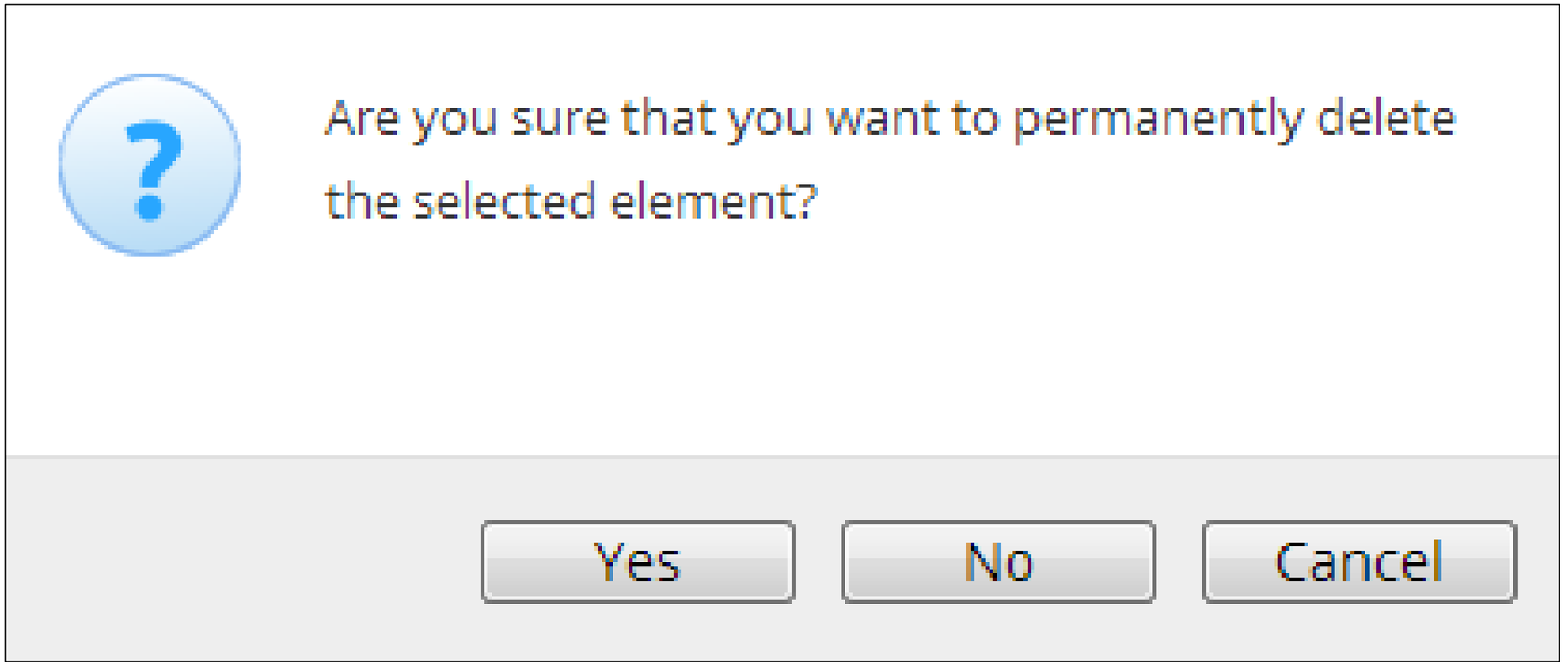


5. Error prevention

Even better than good error messages is a careful design which prevents a problem from occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.

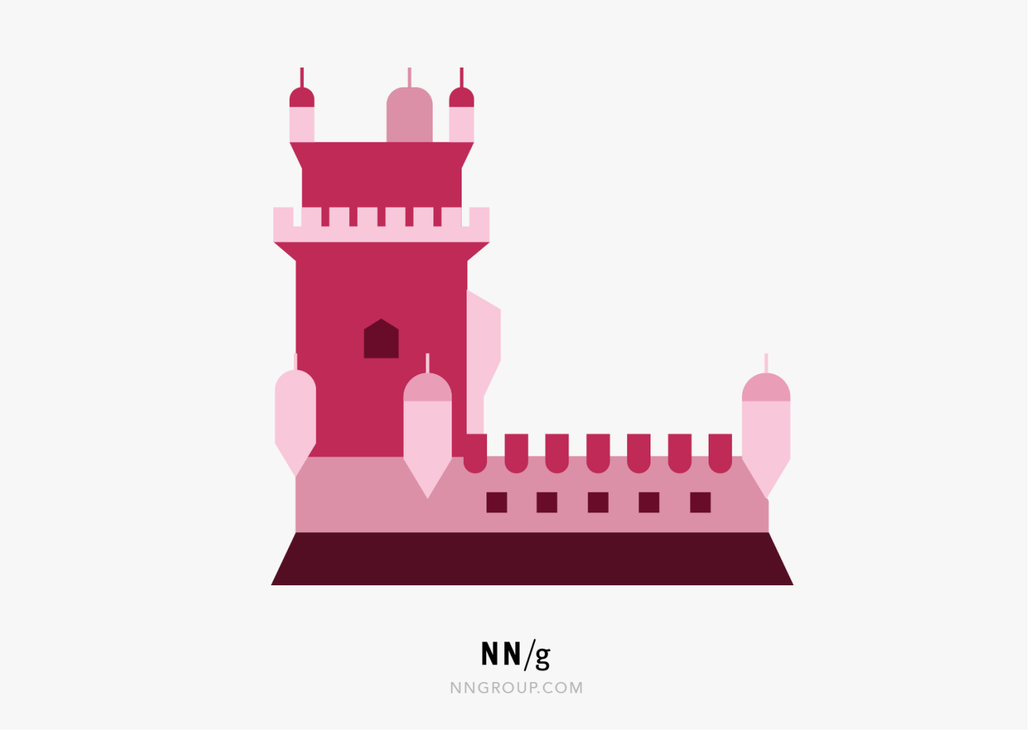


5. Error prevention

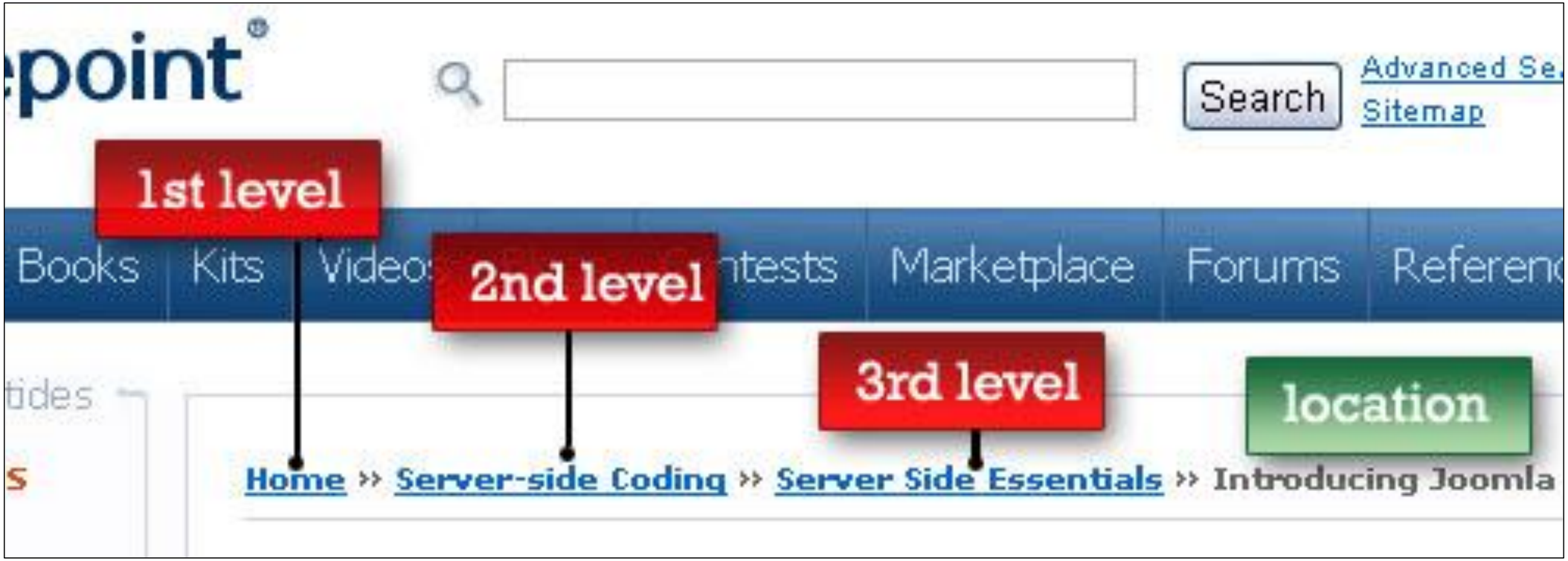


6. Recognition rather than recall

Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.

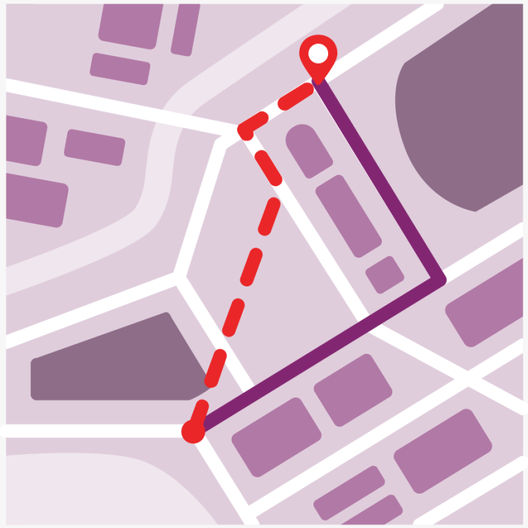


6. Recognition rather than recall



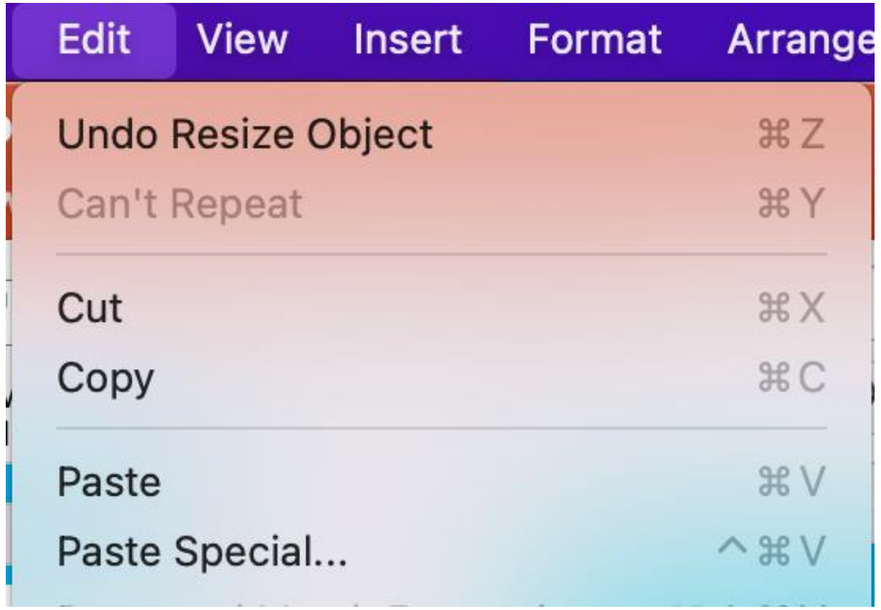
7. Flexibility and efficiency of use

Accelerators – affordances or interaction techniques unseen by the novice user – may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users.
Allow users to customize frequent actions.



7. Flexibility and efficiency of use

CTRL-C &
CTRL-V

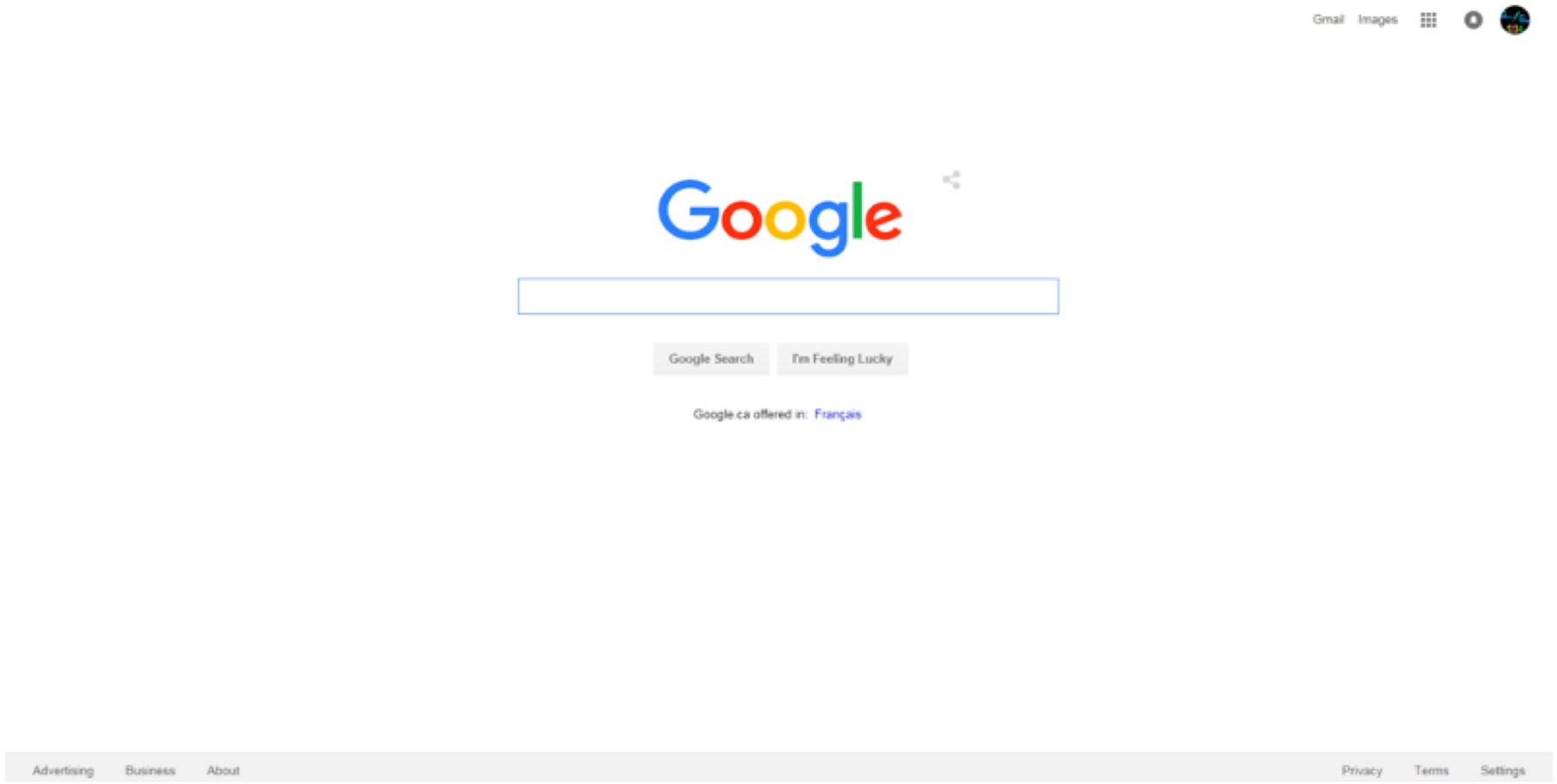


8. Aesthetic and minimalist design

Interfaces should not contain information which is irrelevant or rarely needed. Every extra unit of information in an interface competes with relevant units of information and diminishes their relative visibility.

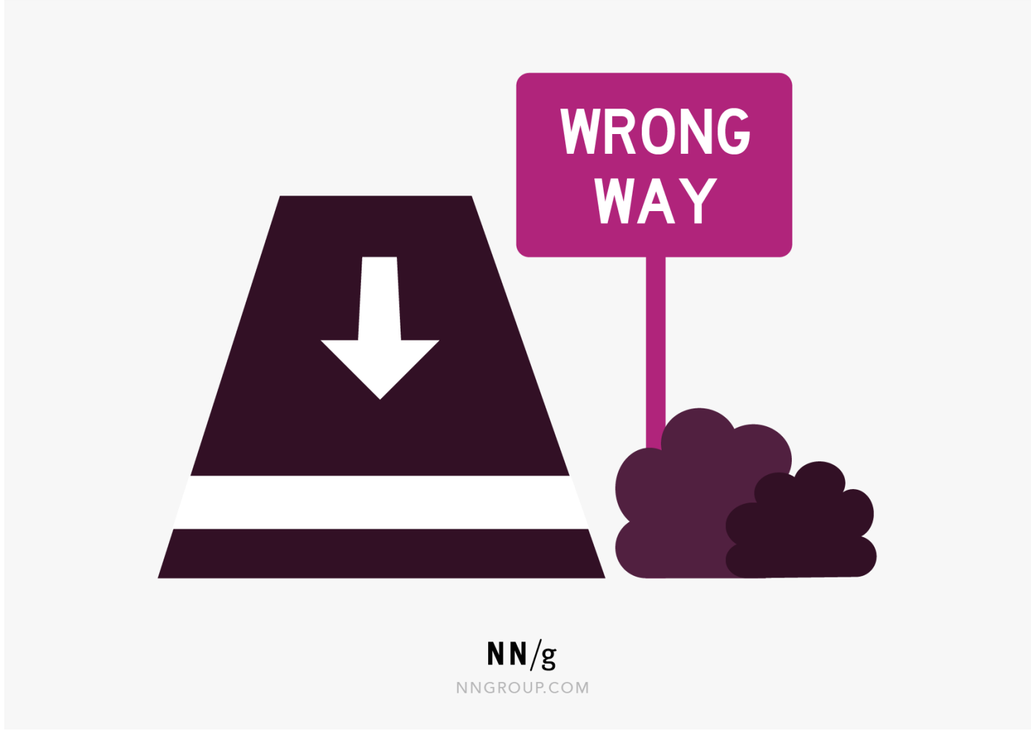


8. Aesthetic and minimalist design

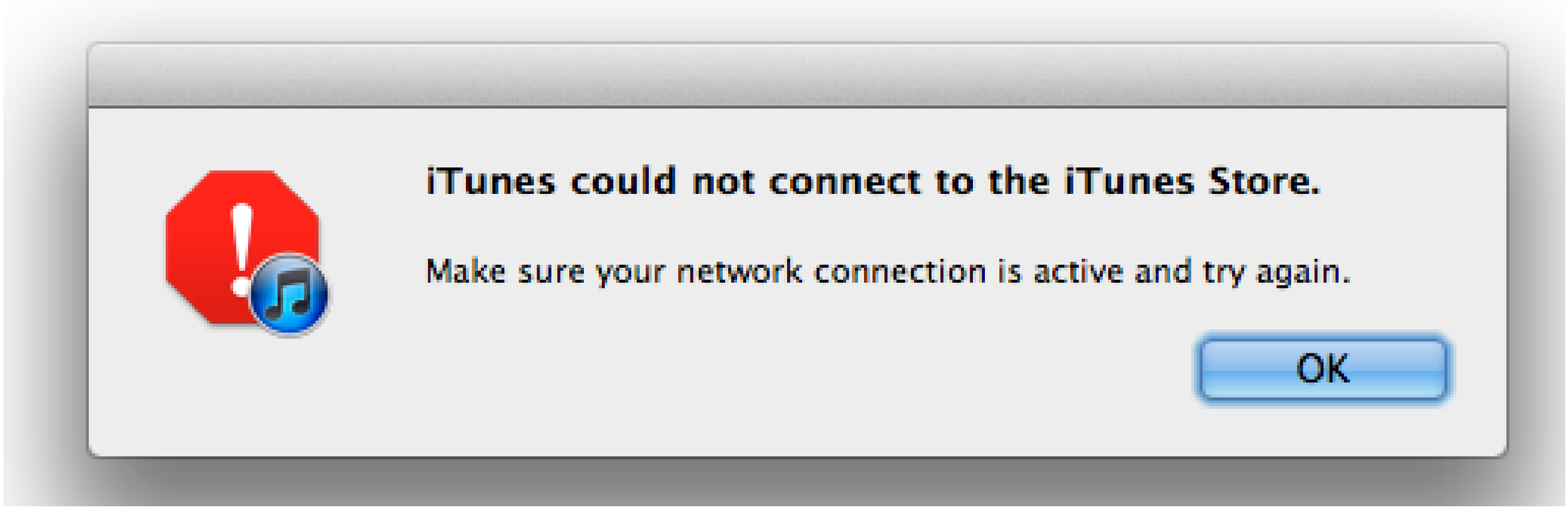


9. Help users recognize, diagnose, & recover from errors

Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

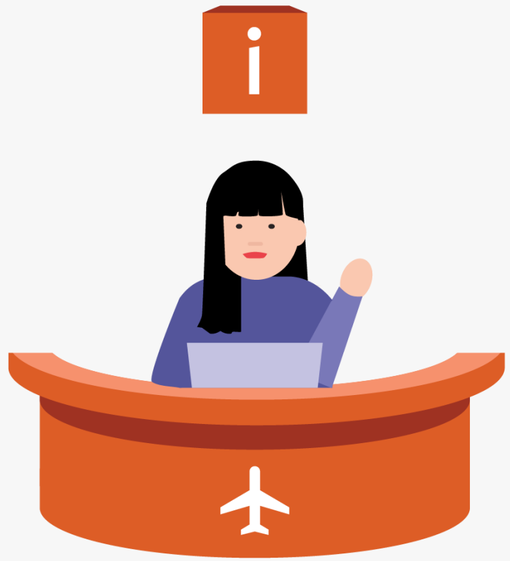


9. Help users recognize, diagnose, & recover from errors



10. Help and documentation

Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.



10. Help and documentation

What do you need help with?

A. I am having trouble reading the screen

B. I need to adjust the volume

C. I do not know how to proceed

D. I want to quit the program

Usability Heuristics

In your A3 report, **use** and **document** how you apply these heuristics.

Source and further information:
https://www.nngroup.com/articles/ten-usability-heuristics/



User Evaluation for A3

For A3, we need to use

one **observation** technique
one **query** technique

Overview

- Observational methods:
  - Think-aloud
  - Cooperative evaluation
  - Post-task walkthroughs
- Query Techniques
  - Interviews
  - Questionnaires



Observation techniques



Think-Aloud

- Participants **verbalise** their thoughts

1. Find a representative set of users
2. Ask them to complete a number of representative tasks
3. Let user complete the task (alone!)

Make your participant feel comfortable — "*Do not be afraid to make mistakes, as they are a display of the system's limitations, not of your ability.*"

Do not provide solutions immediately if something goes wrong.
Let the user resolve.

Cooperative Evaluation

- Analyst and user evaluate the system **together**

Make participant feel like a product owner – "*We very much appreciate your input and will use your suggestions to improve our product to more closely match your expectations.*"

Discuss not only the identified problems, but also ask users to propose solutions which they think will work for them.

Post-task walkthrough

- Users' use of the system is recorded (on-screen, video taped, written down, etc.)
- Analyst asks each user to **reflect** on certain events during their use of the product, **after** task completion.

Make sure to introduce the context of the situation you are discussing with your participant.

Query Techniques



Interviews

- One-to-one conversation between a user and the analyst.

Prepare questions, but do not be afraid to discuss relevant topics as they come up during conversation. Keep it a human conversation, not completing a list of questions in an automated mode.

"*Can you explain this further?*"

Questionnaires

- List of questions given to user (usually after completion of a set of tasks)
- Validated questionnaires (e.g., SUS – System Usability Scale)
- Allows for quantitative analysis

Come do research with us!

- Investigating online **conformity**
- Investigating online **misinformation**
- Investigating **Virtual AI influencers**
- User perceptions of different **Artificial Intelligence**
- **Twitter** data scraping and analysis
- Improving **crowdsourcing** data using worker attributes
- **Any other projects** you propose!



Contact us at *jorge.goncalves@unimelb.edu.au*

Next Class

Final Feedback Session

10	Oct-05	L10: Assignment 3 Feedback	T9: How to Make a Good Demo
11	Oct-12	L11: Assignment 3 Feedback	T10: Evaluation
12	Oct-19	Assignment 3 Demos	T11: Assignment 3 Support & Feedback

Questions?

Post them on Ed!